

Engineering Innovation for Sustainable Futures



This year, student teams are invited to respond to the question:

“How can engineering innovation be used to create business opportunities that address today’s pressing social and environmental challenges?”

Your project may focus on one of the following areas:

1. Sustainable Mobility and Future Transport
2. Net-Zero Aerospace and Advanced Manufacturing
3. Renewable Energy and Smart Campus Systems
4. Smart Infrastructure, Robotics, AI and Digital Engineering
5. Inclusive Engineering and Assistive Technologies

Your challenge is to select a specific opportunity within one of these themes and propose:

- an engineering project idea that demonstrates technical feasibility,
- and a business model that shows how this solution can be operationalised and sustained.

The focus is on combining technical depth, commercial awareness, and societal value into a compelling business pitch linking engineering design with an entrepreneurial opportunity.

1. Sustainable Mobility and Future Transport

How can engineering help Bristol and the West of England deliver a cleaner, smarter and more inclusive transport system?

Location of focus: Bristol and the West of England transport network

Examples:

- 1. Autonomous and connected transport systems.*
- 2. Electric vehicle charging infrastructure.*
- 3. Active travel and micromobility solutions.*
- 4. Smart logistics and freight systems.*
- 5. Shared and demand-responsive transport.*
- 6. Lightweight vehicle and transport technologies.*

Why relevant? Bristol faces transport congestion, accessibility challenges, air-quality concerns, and increasing pressure to decarbonise transport systems.

e.g. [https://supersmith.co/compact and foldable mobility scooters](https://supersmith.co/compact-and-foldable-mobility-scooters).

2. Net-Zero Aerospace and Advanced Manufacturing

How can engineering innovation support low-carbon aerospace and sustainable manufacturing?

Location of focus: Filton Enterprise Area and the South West aerospace and manufacturing cluster.

Examples:

- Hydrogen and electric aviation concepts.*
- Lightweight aircraft structures and advanced composites.*
- Sustainable and circular manufacturing systems.*
- Product life-extension, repair, and remanufacturing.*
- Digital manufacturing and Industry 5.0 technologies.*
- Industrial decarbonisation and resource efficiency.*

Why relevant: Filton Enterprise area is home to one of the UK's largest aerospace and advanced engineering clusters, creating opportunities for innovation in sustainable aviation and manufacturing.

e.g. [HyPRIME](#) hydrogen infrastructure for the future of commercial aviation.

3. Renewable Energy and Smart Campus Systems

How can we design affordable, resilient, and sustainable energy systems for campuses, communities, and businesses?

Location of focus: UWE Bristol Frenchay Campus.

Examples:

- Solar energy generation and battery storage.*
- Smart buildings and energy management systems.*
- Smart grids and microgrids.*
- Renewable energy integration.*
- Electric vehicle charging and vehicle-to-grid systems.*
- Water, energy, and resource monitoring technologies.*

Why relevant: Universities and communities must reduce carbon emissions while improving energy resilience, affordability, and operational efficiency.

e.g. [Heat networks](#) - decarbonising heating systems.

4. Smart Infrastructure, Robotics, AI and Digital Engineering

How can intelligent technologies improve the sustainability, resilience, and performance of infrastructure and essential services?

Location of focus: Bristol and the South West's infrastructure and technology sectors.

Examples:

- *Robotics for infrastructure inspection and maintenance.*
- *Drones for surveying and environmental monitoring.*
- *Digital twins for buildings, transport, and infrastructure assets.*
- *AI-enabled predictive maintenance.*
- *Smart sensing technologies for water, energy, and waste systems.*
- *Data-driven decision-support tools for councils, businesses, and infrastructure operators.*

Why relevant: The region faces challenges associated with ageing infrastructure, climate adaptation, resource management, and digital transformation.

e.g. [Robotiz3d](#) - AI-enabled autonomous robotic systems for infrastructure inspection and predictive road maintenance.

5. Inclusive Engineering and Assistive Technologies

How can engineering solutions improve accessibility, independence, wellbeing, and quality of life for diverse users?

Location of focus: UWE Bristol, Bristol communities, and regional healthcare providers.

Examples:

- *Mobility aids and adaptive technologies.*
- *Assistive robotics.*
- *Wearable health and rehabilitation technologies.*
- *Inclusive transport solutions.*
- *Accessible digital services.*
- *Smart navigation systems.*
- *Technologies supporting independent living.*

Why relevant: An ageing population and increasingly diverse communities require innovative engineering solutions that improve accessibility, inclusion, and wellbeing.

e.g. [Open Bionics](#) - affordable, 3D-printed prosthetic arms.

Please note that your project has the following conditions.

- **Budget: max £500,000**
- **Duration: max 2 years**

Resources:

[Bristol One City](#)

[Filton Enterprise](#)

[Sustainability at UWE Bristol](#)

[Creating a greener future](#)

[Sustainable accommodation at UWE Bristol](#)

[South West England is at the forefront of cyber security, green tech, aerospace....](#)